6.2. Different views of superficiality inform different regions of the Pareto curve

The estimated (b, δ) Pareto curves in Figure 3 contain points from various methods corresponding to the different views of superficiality described in Section 2. Figure 1 shows how each of these views contributes to a particular region of the Pareto curve (plots for other models and tasks are provided in Figure 6). At the shortest program lengths, we see that the Pareto curve is mostly informed by the ③ inference-control view. These methods (URIAL and ICL) bring modest gains in performance with minimal program size. In the medium regimes of program lengths, we observe that the ① data view dominates. This method (Subset Training) improves performance over the the inference-control methods, but at the cost of encoding much more data in the program P. Finally, the longest programs always correspond to the ② parameter view (LoRA and Bayesian-LoRA), whose methods directly encode parameters in P. While the three views supporting the SAH (①, ② and ③) seemed orthogonal at first, they are now thus directly compared on a common, quantitative footing. Instead of competing explanations, they are rather important and complementary techniques to estimate the length–performance (b, δ) Pareto curves.

6.3. Revisiting previous claims about superficiality

In light of the results discussed above, we now revisit previous claims made about the superficiality of LLM adaptation.

Raghavendra et al. (2024): Fine-tuning on few samples does not saturate performance. The original formulation of the SAH left unclear what 'knowledge' and 'capabilities' implied Both Raghavendra et al. (2024) and Lambert (2025) interpret it as implying that fine-tuning on few data points should quickly saturate performance on a task; they then found that this is generally not the case. We intentionally incorporated the target performance level δ in our definition of SAH (Definition 3.6), as some tasks may require arbitrarily large amounts of information to achieve particularly high performance levels. For example, Olmo3 32B can be adapted to achieve 78.4% in GSM8K with a program as small as a single ImageNet image. However, we are not able to substantially improve upon these results even in our largest programs, suggesting that perhaps considerably more information is needed in order to saturate the task.

Liu et al. (2024): Fine-tuning updates can be worth as little as 1 bit per parameter. There is a growing interest in compressing the weight difference between pre-trained and fine-tuned model, which is sometimes termed delta compression (Ping et al., 2024; Huang et al., 2025; Yadav et al., 2025). They are similarly motivated by the intuition that fine-tuning adds less information to a model than pre-training, and therefore it may be possible to compress this update. Liu et al. (2024), for example, compress the difference between a pre-trained and fine-tuned model to approximately 1 bit per parameter, with minimal performance impacts on their analysed tasks. They interpret these results as indicative of the low amount of information gained during fine-tuning. Under our framework, these results understate the amount of information that fine-tuning adds to a model. For example, the size of a full fine-tuning dataset is much smaller than the size of 1-bit per parameter of an LLM. Comparing our smallest model (SmolLM3) to our largest dataset (GSM8K), the dataset size is 3×10^7 bits while the model has 3×10^9 parameters. These methods, thus, would produce programs P that are neither small nor performant enough to be Pareto optimal under our operationalisations.

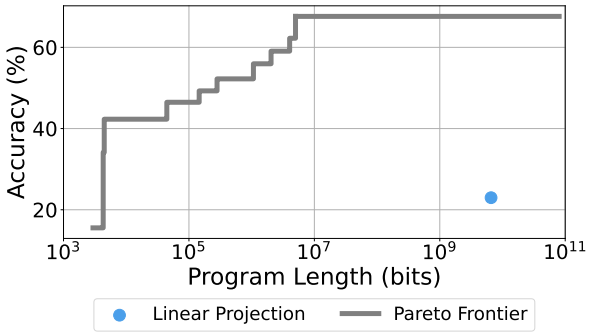

Figure 4. Performance of the linear projection fine-tuning proposed by Chen et al. (2025) in comparison to our estimated Pareto curve. The performance of the method is far from the Pareto curve.

Chen et al. (2025): Fine-tuning on reasoning tasks is not superficial. Chen et al. (2025) critique the SAH for not rigorously defining what 'superficial knowledge' means. They then define superficial knowledge as the performance a pre-trained model can achieve when only adapting the linear projection of the output layer. This is based on the intuition that if learning a linear transformation on top of a pre-trained model is sufficient for performing well on a task, then most of the knowledge required for the task is already present in the pre-trained model itself. Empirically, they show that reasoning tasks such as GSM8K require more than superficial knowledge, because fine-tuning only the output layer is not sufficient for achieving good performance. Our framework reveals that their approach might not be optimal for finding short adaptation programs. In Figure 4, we show the length–performance of their method compared to the Pareto curve when adapting OLMo3-7B to GSM8K. This figure illustrates that training only the output layer leads to neither short programs nor high accuracies (see similar results for other tasks in Figure 7 in Appendix F). More broadly, Chen et al.'s approach is closely related to probing, and—as argued by prior work in that field (Hewitt & Liang, 2019; Pimentel et al., 2020b)—we see no reason why 'superficiality' should be restricted to linearly encoded knowledge.

7. How Does Pre- and Post-training Affect the Superficiality of Adaptation?

Finally, to demonstrate the usefulness of our definitions beyond revisiting previous claims about superficiality, we now show that (b, δ) Pareto curves can shed light on how post-training affects the superficiality of adaptation.

Experimental setup. We compute (b, δ) Pareto curves for mathematical reasoning (GSM8K) and instruction following (IFEval) using randomly initialised models as well as the final pre- and post-training[8] checkpoints of SmolLM3 3B and Olmo3 7B (results for machine translation are available in Figure 8 in Appendix F). We use the same evaluation setup, including the same minimal prompting templates, for all these checkpoints. Figure 5 shows the Pareto curves obtained when running our methods on these models θ.

Pre-training makes strong performance accessible. As expected, in this figure, we see that pre-training enables the model to reach much higher task performance compared to randomly initialised checkpoints. For example, in the case of Olmo3-7B, the best performance we are able to achieve on mathematical reasoning (GSM8K) improves from 1.1% to 67.6%, and on instruction following (IFEval) from 26.6% to 68.5%. Further, this figure also shows that the allowed program size significantly affects the performance of pre-trained checkpoints. For example, with programs of 10^5 bits, the performance obtained in each setting is still far below its maximum performance. Finally, at the maximum achieved performance levels δ, the task complexity is relatively high; most evidently in SmolLM3 on GSM8K, performance only plateaus at 5×10^{10} bits (around 6.2 GB).

Post-training collapses the complexity of achieving strong performance. As expected, post-training further increases task performance, e.g., from 58.4% to 82.8% in mathematical reasoning and from 49.6% to 70.5% for instruction following for SmolLM3.[9] More importantly, however, post-training collapses the complexity of achieving the maximum performance. For example, for pre-trained Olmo3-7B the complexity of achieving its maximum performance is 5×10^6 bits for GSM8K, and 2.5×10^7 bits for IFEval. In contrast, for post-trained models, program size has minimal effect on performance beyond a size of 10^4 bits. These results provide an information-theoretic interpretation of what it means for post-training to resurface what was learned in pre-training: the complexity of accessing strong performance is substantially reduced.

[8] See Appendix E for details on the data used for pre- and post-training these models.

[9] We note that the training data for both GSM8K and IFEval were included in the post-training of both SmolLM3 and Olmo3 (see Appendix E), which could partially explain this effect.

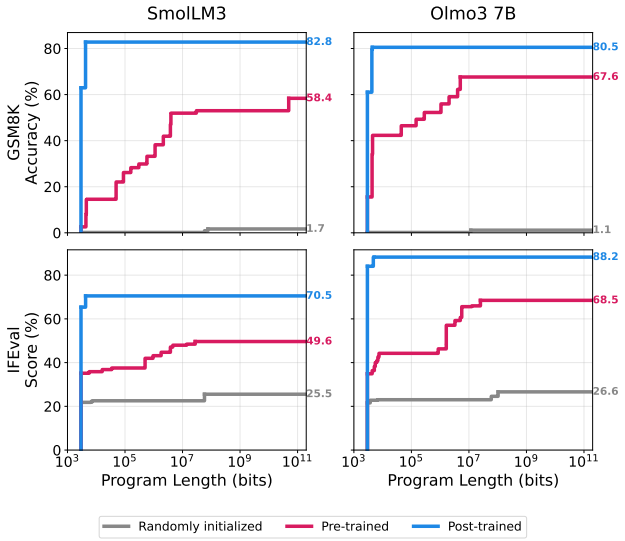

Figure 5. Estimated complexity curves conditioned on pre- and post-trained checkpoints of SmolLM3 and Olmo3 for math (GSM8K) and instruction (IFEval) tasks. Pre-training allows access of strong performance in these tasks. Post-training improves these performances, but more importantly, it greatly reduces the complexity to reach such performance.

8. Conclusion

The superficial alignment hypothesis has lacked a precise definition, which led both to seemingly orthogonal interpretations of how to support this hypothesis and to important critiques against it. We address this here by defining task complexity—the length of the shortest program that achieves a target performance on a task—which we use to formalise the SAH as claiming that access to a pre-trained model drastically reduces the complexity of achieving high performance on many tasks that would otherwise be highly complex. Our framework unifies previous arguments in support of the SAH (the data view, the parametric view, and the inference control view, see Section 2) as proposing different strategies for finding short, high-performance programs for a target task. Experimentally, we estimate the task complexity of mathematical reasoning, machine translation, and instruction following, conditioning these complexities on three different models. We find that programs as small as a few kilobytes can already adapt pre-trained models to achieve strong performance on these tasks. By analysing intermediate training checkpoints of LLMs, we show that pre-training enables access to strong performance, though often at high complexity (requiring megabytes to gigabytes long programs), while post-training dramatically collapses this complexity, making high performances accessible with less than a kilobyte. We believe this to be an important step towards understanding the concept of superficiality in LLMs, and the information contained in them.

9. Limitations and Future Work

Our operationalisation of the SAH inherits fundamental properties of algorithmic information theory that shape and limit what claims we can make. First, because task complexity is uncomputable (Corollary A.5), our empirical estimates provide only upper bounds. Therefore, our (b, δ) Pareto curves show which performance levels are achievable with different program lengths, but they are not provably tight, as there may exist shorter programs we have not discovered. Second, while we can reasonably upper-bound the conditional task complexity $\mathrm{C}(\mathtt{T}_\delta \mid \theta)$ using the three views on superficiality, estimating the task complexity $\mathrm{C}(\mathtt{T}_\delta)$ is difficult, as it requires finding the shortest program to solve $\mathtt{T}_\delta$ without access to any pre-trained model. This prevents us from directly measuring the information $\mathrm{I}(\mathtt{T}_\delta; \theta)$ (see Definition 3.5), which is why our formalisation of the SAH (Definition 3.7) relies solely on conditional complexity. We hope future work can explore the tightness of our approximations.

Beyond the SAH, task complexity may offer useful perspectives on other problems in ML. For instance, work on quantifying the safety of open-weight model releases (Wallace et al., 2025; Che et al., 2025) deals with measuring how easily unsafe capabilities can be accessed. Our framework naturally operationalises this as $\mathrm{I}(\mathtt{T}_\delta; \theta)$ for some unsafe task $\mathtt{T}$ and performance δ: if low, releasing the model weights does not make $\mathtt{T}_\delta$ easier to solve.

10. Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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